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Operando Spectroscopic Identification of Active Sites in NiFe Prussian Blue Analogues as Electrocatalysts: Activation of Oxygen Atoms for Oxygen Evolution Reaction

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ABSTRACT: Developing highly efficient oxygen evolution reaction (OER) catalysts and understanding their activity are pivotal for electrochemical conversion technologies. Here, we report NiFe Prussian blue analogue (PBA) as a promising electrocatalyst for OER in alkaline conditions. This material has an impressively low overpotential of 258 mV that reaches a current density of 10 mA cm⁻². Postmortem characterization showed that the as-prepared catalyst is entirely transformed into amorphous nickel hydroxide after the electrochemical treatment, and the Ni(OH)₂ acts as the active species. Operando X-ray spectroscopic studies further found that this *in situ* generated Ni(OH)₂ displays a unique feature that allows deprotonation under applied potential creating NiOOH_{2-x} that contains Ni⁴⁺ ions. The deprotonation reaction is reversible and potential-dependent, i.e., the amount of Ni⁴⁺ increases with increasing applied potential. Theoretical calculations were used to show that the role of Ni⁴⁺ is to trigger oxidized oxygen ions as electrophilic centers with the subsequent activation of anion redox reactions for OER.

INTRODUCTION

The oxygen evolution reaction (OER) is crucial for renewable energy technologies, including water splitting and rechargeable metal/air batteries¹⁻³. However, its intrinsically sluggish reaction kinetics requires active and cost-effective heterogeneous catalysts. The usage of alkaline electrolytes allows transition metal (TM) compounds as catalysts to replace the common, yet expensive, noble metal oxides⁴⁻⁵. Many low-cost 3d transition metal (oxy)-hydroxides⁶⁻¹⁰, oxides¹¹⁻¹⁷, sulfides¹⁸, selenides¹⁹⁻²⁰ and phosphides²¹⁻²² have been developed as candidate electrocatalysts.

While attempting to rationally design OER electrocatalysts, great effort has been spent on identifying the physical origin of the OER activity—this could help guide the design of new catalysts. An understanding of OER catalysts was previously based on the assumption of a well-defined electrode/electrolyte interface where intermediate species adsorb to the surface metal sites for oxygen evolution. Consequently, the electronic configuration of the metal ions was used as a descriptor—the surface oxygen should bind neither too strongly nor too weakly according to Sabatier's principle²³⁻²⁴. Nevertheless, the latest research results showed that catalysts that undergo OER often have reconstructed surfaces. For instance, Fabbri et al. recently unraveled a dynamic surface self-reconstruction of the well-known Ba_{0.5}Sr_{0.5}Co_{0.8}Fe_{0.2}O_{3-δ} catalyst that led to a (Co/Fe)O(OH) active layer²⁵. Similar data were seen with Co₃O₄²⁶ and CoO_x²⁷ systems. In particular, the discovery of a retained self-reconstructed layer after OER illustrates that metal (oxy)-hydroxides are the actual catalysts. Therefore, the underlying mechanism of oxides for OER should be similar to

(oxy)-hydroxides. Moreover, these examples highlight the need to study OER catalysts under the operando condition.

Recently, metal organic framework (MOF)-derived materials have been extensively studied as a new class of electrocatalysts for OER²⁸⁻³². One cheaper alternative is Prussian blue analogues (PBAs), which exhibit promising electrocatalytic activity³³⁻³⁵. The conventional understanding is that the high reactivity of these systems is due to their high surface area and uniform porosity, which offer many active sites. On the other hand, Zhao et al. found that coordinatively unsaturated surface atoms in ultrathin MOF sheets play a crucial role in OER activity by providing open sites for adsorption²⁸. These explanations assumed that these materials have well-defined chemical structures and readily accessible active sites under electrochemical conditions. Hence, the observation of excellent reactivity in these materials raised a question, that is, whether there is an intrinsic reaction mechanism differing from that of oxides. In this respect, operando studies on these systems are rare relative to oxides and (oxy)-hydroxides.

Herein, we synthesized a new OER catalyst NiFe-PBA (denoted by NF-PBA) that exhibits electrocatalytic activity superior to the state-of-the-art noble metal catalyst IrO₂. We then used it as a model to explore the origin of the activity at the atomic level. Operando X-ray spectroscopy (XAS) combined with *ex situ* characterization demonstrated that the amorphous nickel hydroxide after OER acts as an active species. More importantly, there is a reversible potential-dependent deprotonation reaction of this *in situ* generated Ni(OH)₂. This creates Ni⁴⁺ ions to activate oxidized oxygen ions as electrophilic centers.

RESULTS AND DISCUSSION

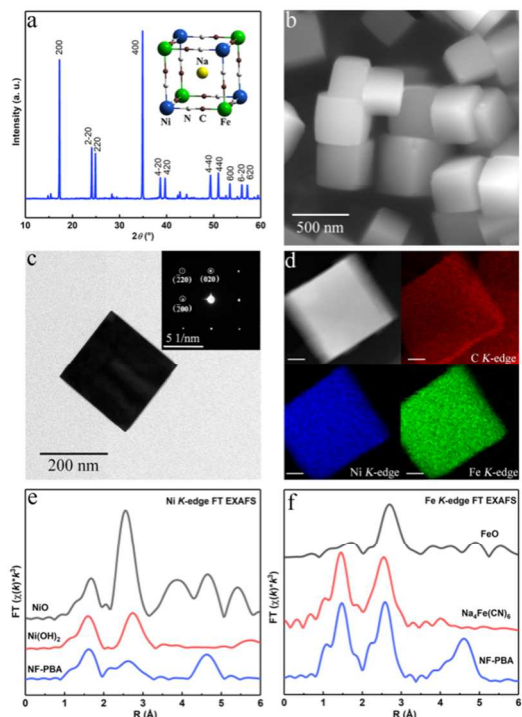


Figure 1. Morphological and structural characterizations of NF-PBA catalysts. (a) XRD pattern of NF-PBA. (b) SEM image of NF-PBA. (c) TEM image of NF-PBA; the inset is the electron diffraction pattern. (d) EDX elemental mapping images of NF-PBA. (e) FT-EXAFS of Ni in NF-PBA. (f) FT-EXAFS of Fe in NF-PBA.

We used a facile wet chemistry method to prepare the NF-PBA nanoparticles, and X-ray diffraction (XRD) characterized the crystal structure of the as-prepared catalyst. The diffraction peaks of NF-PBA in Figure 1a agreed well with previous NiFe PBA structures³⁶⁻³⁷. The scanning electron microscopy (SEM) image in Figure 1b showed that the catalyst has a cubic morphology with dimensions of about 200 nm. This was confirmed with transmission electron microscopy (TEM) (Figure 1c). The inset of Figure 1c showed fast Fourier transform (FFT) analysis of the high resolution TEM image, including (020), (-200), and (-220) faces. These confirm the cubic structure of NF-PBA. The energy-dispersive X-ray (EDX) mapping revealed homogeneous distribution of Fe, Ni and C elements throughout the catalyst structures (Figure 1d). To investigate the electronic and local coordination structures of NF-PBA, we performed X-ray absorption spectroscopy (XAS) measurement. The Fourier transform extended X-ray absorption fine structure (FT-EXAFS) spectra of NF-PBA (Figure 1e and Figure S1) identified a six-fold coordination structure of Ni with the nearest Ni-N bond length of 2.08 Å; Fe was coordinated with six carbon atoms for an average Fe-C bond length of 1.90 Å (Figure 1f and Figure S3). Soft X-ray absorption spectroscopy (s-XAS) determined the oxidation and spin states of transition metals. The Ni L-edge spectra had the same features as NiO, suggesting that the Ni²⁺ oxidation state has a high-spin configuration in NF-PBA (Figure S2). The Fe in NF-PBA was low-spin Fe²⁺ (Figure S4). These results indicated a

well-defined cubic structure in the NF-PBA catalyst as prepared.

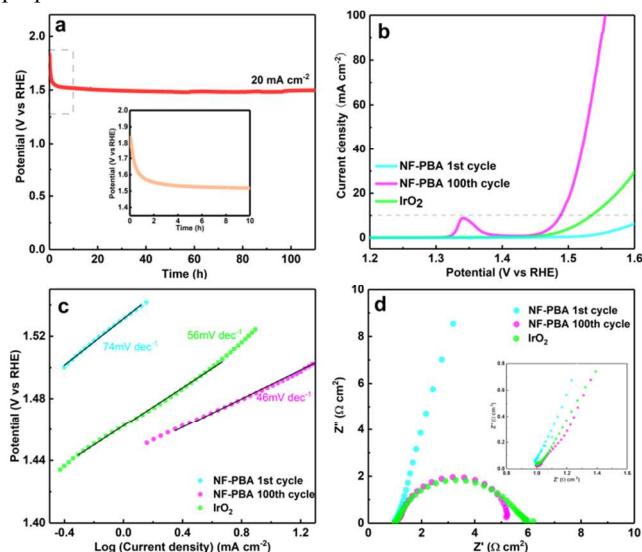


Figure 2. Electrochemical characterization of NF-PBA catalysts. (a) Chronopotentiometric measurement at $j = 20 \text{ mA cm}^{-2}$ for NF-PBA. (b) Polarization curves for NF-PBA in 1 M KOH for different cycles and IrO₂ on the glassy carbon electrodes. (c) Tafel plots of the polarization curves in (b). (d) EIS recorded at 1.488 V (versus RHE), the inset shows data in the high frequency region.

After characterizing the pristine catalyst, we investigated the electrocatalytic OER activities of NF-PBA in 1 M KOH solution. Chronopotentiometric measurements at a constant current density of 20 mA cm^{-2} indicated an activation process for the as-prepared catalyst (Figure 2a). For instance, the required potential dramatically decreased over the first 10 hours and then maintained nearly a constant value for 100 hours demonstrating excellent durability for the fully activated catalyst (denoted by NF-PBA-A). This activation process also can be reflected by the cyclic voltammetry measurement, as shown in Figure S5. Linear sweep voltammetry (LSV) measurement (Figure 2b and Figure S6) showed that the initial catalyst has poor OER activity. By contrast, the NF-PBA-A sample has an impressively low overpotential of only 258 mV to reach a current density of 10 mA cm^{-2} . The current density on NF-PBA-A at an overpotential of 304 mV can reach 61 mA cm^{-2} , which is about 6 times higher than the noble metal catalyst IrO₂ (10 mA cm^{-2}). The decreased Tafel slope of 46 mV dec^{-1} suggested significantly enhanced activity of NF-PBA upon electrochemical treatment (Figure 2c and Figure S7). The charge transport kinetics was also strengthened by the activation. The electrochemical impedance spectroscopy (EIS) data in Figure 2d indicated that the activated NF-PBA-A exhibited much lower charge transfer resistance (R_{ct}) than pristine NF-PBA (Figure S8 and Table S3). Overall, these electrochemical results demonstrated that the NF-PBA catalyst underwent an activation process during OER implying a transformation of the catalyst structures that contribute to its excellent OER performance.

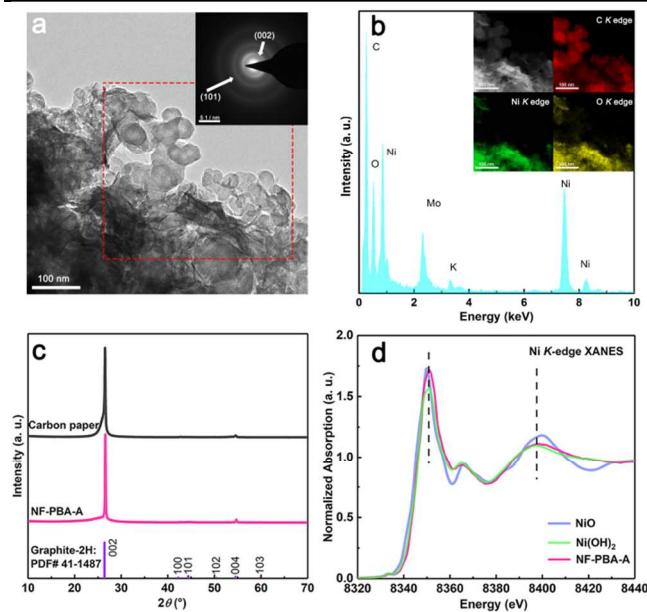


Figure 3. Morphological, geometric, and electronic structural characterizations of NF-PBA-A. (a) TEM image of NF-PBA-A; the inset is the electron diffraction pattern. (b) EDS and elemental mapping images of NF-PBA-A. (c) XRD pattern of NF-PBA-A. (d) Ni K-edge XANES spectra of NF-PBA-A compared to NiO and Ni(OH)₂.

We next performed systematic postmortem investigations of the fully activated catalyst NF-PBA-A to understand how the electrochemical treatment promotes OER activity. The SEM image (Figure S9) and the TEM image (Figure 3a) indicated that initial cubic morphology of NF-PBA transformed into shapeless morphology. However, the carbon black remained spherical shape. The elemental mapping results exhibited a distribution of Ni and O elements of remaining catalyst while almost no Fe and N signals were detected. This observation was further supported by the energy-dispersive spectroscopy (EDS) data (Figure 3b). The SAED pattern only exhibited (002) and (101) rings, which were both attributed to the carbon black (inset of Figure 3a). This illustrates that the catalyst had profound change while carbon black remained unchanged. The amorphous nature of the NF-PBA-A was confirmed via the XRD pattern that exhibited only peaks from the carbon paper (Figure 3c and Figure S10); no signal of NF-PBA was seen. These findings suggest that the pristine NF-PBA structure completely degraded into a new amorphous structure containing Ni and O under electrochemical treatment.

We also used X-ray spectroscopy to identify the local and electronic structure of the *in situ* generated active species. X-ray absorption near edge structure (XANES) was performed on the Ni K-edge of NF-PBA-A as well as NiO and Ni(OH)₂ references (Figure 3d). The XANES structures of NF-PBA-A were consistent with that of Ni(OH)₂, whereas distinctly different from that of NiO, revealing that NF-PBA-A has an analogous structure of Ni(OH)₂. This result was further supported by O K-edge XAS data (Figure S11), which exhibited an obvious feature at 533.5 eV arising from hydroxy species. The quantitative structural information was obtained by fitting the FT-EXAFS data (Figure S12 and Table S1). The bond length and coordination number of the first Ni-O shell

were 2.04 Å and 6.0, respectively. Moreover, the bond length and coordination number of the Ni-Ni shell were 3.08 Å and 4.6, respectively. Combined with morphologic and spectroscopic characterizations, we demonstrated that the pristine NF-PBA catalyst was entirely transformed into an amorphous Ni(OH)₂ structure after OER activation in alkaline media.

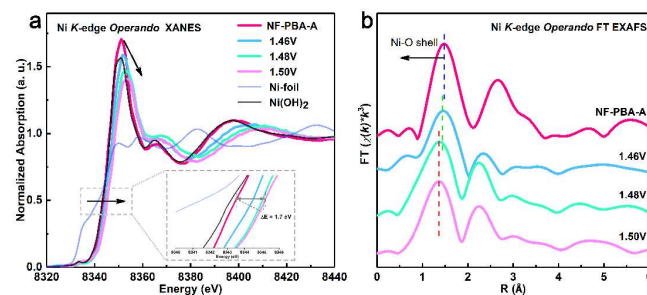


Figure 4. Operando Ni K-edge XAS spectra of NF-PBA-A under different potentials. (a) XANES of NF-PBA-A as well as references. The inset shows the shift of Ni K-edge position. (b) FT-EXAFS of NF-PBA-A.

Recent operando studies of the evolution of catalysts were considered to be critical to understand the intrinsic catalytic mechanism of electrochemical water splitting compared to *ex situ* measurements. The initial NF-PBA structure completely changed to amorphous nickel hydroxide. Thus, further study on this fully activated species by operando techniques is the key to understanding the underlying mechanism for OER. Therefore, we conducted operando Ni K-edge XAS studies to track the evolution process of the geometric and electronic structures of the NF-PBA-A catalyst as a function of the applied potential (Figure 4). The energy position of the absorption edge gradually shifted to higher energy with increasing potential, i.e., the spectra showed a distinct edge shift of 1.7 eV under catalytic potential of 1.50 V. This suggests oxidation of Ni²⁺ ions to higher valence states (Figure S13). More importantly, this potential-dependent process is reversible (Figure S14). These occurrences indicate that the Ni(OH)₂ generated *in situ* is the actual catalyst. Attempting to further explore the difference between amorphous Ni(OH)₂ and its crystalline counterpart, we also synthesized crystalline Ni(OH)₂. The detailed characterizations can be seen in Figure S15 and S16. The operando XAS experiment on this sample showed a smaller change in Ni valence state upon the applied potential at 1.50 V, as shown in Figure S17. This illustrates that the nickel ions in amorphous system are able to be readily oxidized to high valence and thus exhibit more reactivity.

The bond length of the first metal-oxygen shell is characteristic of the oxidation state of metal ions—especially in studies of cathode materials from Li-ion batteries³⁸. Therefore, we performed FT-EXAFS analysis to illustrate the local structure of Ni during OER (Figure S18 and Table S4). In the initial NF-PBA-A structure, the average Ni-O bond length was 2.04 Å, which indicated that Ni ions were predominantly in a 2+ oxidation state. When a low potential was applied, the Ni-O bond length decreased to 1.99 Å, which suggested a Ni²⁺/Ni³⁺ mixed state. Further oxidation to Ni³⁺/Ni⁴⁺ was confirmed by the shrinking of Ni-O bond length to 1.89 Å at high applied potential. There was no obvious change at higher potential. The FT-EXAFS results showed that the coordination number of the first Ni-O shell remained

nearly constant throughout the process in accordance with the weak pre-edge peaks characteristic of the octahedral configuration. This revealed a well-preserved octahedral structure independent of the applied potential.

Having found the raising of the oxidation, remaining of the coordination number, as well as contraction of the Ni-O bond length, we speculated on a potential-dependent deprotonation reaction scheme (Figure 5a). When the initial NF-PBA-A catalyst (in $\text{Ni}(\text{OH})_2$ structure) interacted with alkaline solution at low potential, partial hydrogen atoms were extracted from the hydroxyl group and a fraction of oxidized Ni^{3+} species began to be present. At higher potential, further oxidation of nickel ions to $\text{Ni}^{3+/4+}$ states occurred by successive deprotonation. These formed partial intermediate $\text{Ni}(\text{O})_2$ species. Further DFT calculations were carried out to validate this hypothesis. The calculated structural parameters are listed in Table S5. The average Ni-O bond length was 2.086 Å in $\text{Ni}^{2+}(\text{OH})_2$. This reduced to 1.969 Å in $\text{Ni}^{3+}\text{OOH}_{1.0}$ and 1.873 Å in Ni^{4+}OO upon deprotonation, which agreed nicely with the EXAFS results and suggested the evolution of nickel valences as a function of applied potential. Thus, we confirmed potential-dependent NiOOH_{2-x} as the active state. This reaction is described in Equation 1.

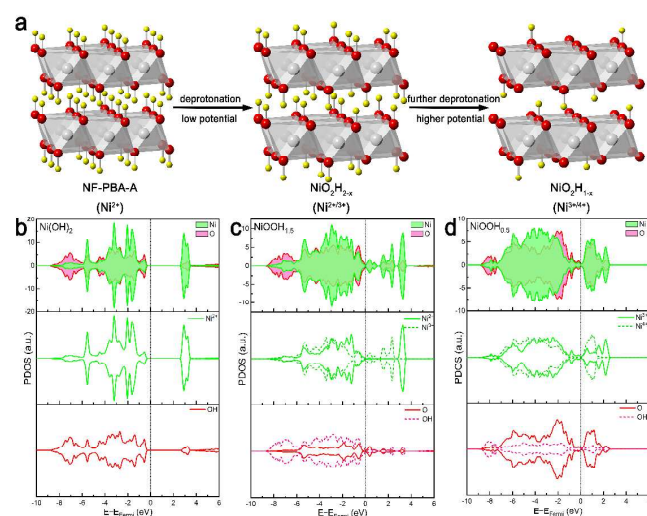
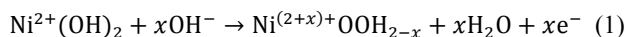


Figure 5. Sketch of the evolution of geometric and electronic structure during OER. (a) Multi-step deprotonation of NF-PBA-A during OER. (b)-(d) Calculated atomic PDOS of $\text{Ni}(\text{OH})_2$, $\text{NiOOH}_{1.5}$ and $\text{NiOOH}_{0.5}$ with different hydrogen content. The Fermi level is at zero. Note that the scale in these systems is different for clarity.

These data showed that the Ni^{4+} ion content increased with increasing applied potential via catalyst deprotonation. This implies that the Ni^{4+} ions have significant contribution to the reactivity for OER. This observation was consistent with recently reported results showing that late transition metals with high oxidation states are among the most active catalysts³⁹⁻⁴¹. To determine how the Ni^{4+} ions promoted OER activity, we plotted the projected density of states (PDOS) obtained by DFT calculations to track the evolution of the catalyst's electronic structure during the deprotonation process (Figure 5b-d and Figure S19). The initial $\text{Ni}(\text{OH})_2$ is an

insulator with a band gap of ~ 2.5 eV. Moreover, the DOS around the Fermi-level was dominated by Ni-3d states. When a small amount of protons was extracted (e.g. $\text{NiOOH}_{1.5}$), the system quickly transferred to a metallic phase. The O 2p component particularly increased in DOS near the Fermi-level. With further increasing degree of deprotonation, the O 2p component near the Fermi-level dramatically raised. Detailed analysis of O 2p PDOS revealed that these increased O 2p states primarily arise from the deprotonated oxygen atoms. This was confirmed by the O 2p band center that gradually shifted to higher energy upon deprotonation (Table S6). More importantly, there were significant O 2p states distributed above the Fermi-level in the system with high degree of deprotonation, indicating the formation of O 2p holes. This finding is analogous to the new observation on the Li-ion battery, i.e., Li^+ removal is charge-compensated by the formation of localized electron holes on O atoms⁴²⁻⁴³. The underlying mechanism is that the Ni^{4+} 3d band deeply inserts into the O 2p band and consequently pushes up the latter to the high energy (Figure S20 and S21). The presence of localized O 2p holes implies that the oxidized oxygen ions have an unpaired electron. These electronic structures are completely in line with the latest proposed anionic redox process-mediated OER mechanism⁴⁴⁻⁴⁶. Therefore, we realized that these oxidized oxygen ions in our system could act as electrophilic centers for OER, and the role of the Ni^{4+} ions played is to activate these oxygen ions. In addition, the high conductivity of operando phase NiOOH_{2-x} should promote the OER activity.

CONCLUSIONS

In summary, we reported a new type of NF-PBA electrocatalyst towards OER in alkaline conditions. Electrochemical measurements showed that the as-prepared catalyst undergoes an activation process during the initial 10 h, and the fully activated sample has an excellent OER activity with an overpotential $\eta = 258$ mV. This produces a constant current over 100 h in chronopotentiometric measurements. Spectroscopic and microscopic studies validated that the catalyst completely degrades into an amorphous $\text{Ni}(\text{OH})_2$ during the activation process. This provides active sites for water oxidation. Operando XAS further found that this *in situ* generated $\text{Ni}(\text{OH})_2$ transformed into a NiOOH_{2-x} structure under applied potential via the deprotonation process. The degree of deprotonation is a function of the applied potential, and this reaction is reversible. DFT calculations showed that the proton removal is charge-compensated by the formation of localized electron holes on O atoms partially coordinated by Ni^{4+} ions. Our work suggested that the materials generated by the *in situ* degradation of molecular frameworks, i.e. transition-metal hydroxides, are active species for OER. This indicates that the underlying mechanism for the materials based on the molecular frameworks is similar to that of the oxides and (oxy)-hydroxides. Importantly, we revealed that the anion redox process induced by the formation of the oxidized oxygen underlies its excellent OER activity.

The finding of high OER activity in $\text{Ni}(\text{OH})_2$ suggested that Ni-based hydroxides should be focused on. In particular, we found that the nickel ions in amorphous system are able to be readily oxidized to high valence and thus exhibit more reactivity compared to those in crystalline system. Overall, we provided an example that a certain material can be completely

dissociated under the OER condition. This suggests a design strategy beyond traditional nano-catalyst design involving nanostructure, support effect and controlling synthesis of nanoparticles. Therefore, it is essential to explore those materials that can readily form hydroxides/(oxy)-hydroxides under the electrochemical condition.

EXPERIMENTAL SECTION

Synthesis of Samples. $\text{Na}_4\text{Fe}(\text{CN})_6 \cdot 10\text{H}_2\text{O}$, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{Na}_3\text{C}_6\text{H}_5\text{O}_7$ were analytical grade and purchased from the Aladdin Company, China. A typical liquid phase method was used to synthesize NF-PBA as reported by previous paper³⁶. Generally, there were two steps. First, 2.119 g $\text{Na}_4\text{Fe}(\text{CN})_6 \cdot 10\text{H}_2\text{O}$ (Aladdin, analytical reagent) was added in 250 ml water to obtain solution A. 5.882 g $\text{Na}_3\text{C}_6\text{H}_5\text{O}_7$ (Aladdin, analytical reagent) and 1 g $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (Aladdin, analytical reagent) were added in another 250 ml water to obtain solution B. Then, the solution A was poured into solution B, and the mixture was aged at room temperature for 48 h to get the precipitate. After that, the as-obtained $\text{Na}_x\text{NiFe}(\text{CN})_6$ nanocubes were filtered, washed with deionized water and dried overnight in vacuum oven at 100 °C. As the reference sample, the crystalline $\text{Ni}(\text{OH})_2$ was synthesized by an aqueous method as previous report⁴⁷.

Electrochemical Characterizations. The electrochemical measurements of all samples (NF-PBA, IrO_2 , $\text{Ni}(\text{OH})_2$, etc.) were carried out under the same test conditions. Typically, 5 mg of electrocatalyst powder and 5 mg of carbon black were dispersed in 1 ml of 1:1 (v/v) deionized-water/isopropanol mix solvent with a 40- μl Nafion solution (5 wt%, Sigma-Aldrich). The mixture was sonicated in an ultrasonic water bath for approximately 1 h to form a homogeneous catalyst ink. Next, 10 μl of the ink was coated onto a glassy carbon disk with a diameter of 5 mm, at a loading rate of approximately 0.25 mg cm^{-2} , and dried at room temperature. The glassy carbon disk electrode was polished with different polishing powders (1.0- and 0.3- μm alpha alumina and 50-nm gamma alumina, in order) and thoroughly cleaned with deionized water and ethanol before loading. All the electrochemical measurements were performed in O_2 -saturated 1 M KOH with a conventional three-electrode on an Auto Lab electrochemical station in which saturated HgO/Hg and platinum wires were used as the reference and counter electrodes, respectively. Linear sweep voltammetry and cyclic voltammetry were obtained at a scan rate of 5 mV s^{-1} and 1 mV s^{-1} , respectively. Electrochemical impedance spectroscopy was run with AC voltage with 10 mV amplitude at a potential of 1.488 V versus RHE within the frequency range 0.1 Hz to 100 KHz. All potentials were corrected to compensate for the effect of solution resistance measured via high-frequency AC impedance. The long-term durability test was performed using chronopotentiometric measurements (CP) with NF-PBA loading amount of 1mg cm^{-2} . The chronopotentiometric curves of the catalysts were collected at a constant current density of 20 mA cm^{-2} .

Structural Characterizations. The crystalline phase of the NF-PBA catalysts before and after the CP test were recorded by grazing incidence X-ray diffraction on a D8-Advance Bruker-AXS diffractometer with an incidence angle of 1° using $\text{Cu K}\alpha$ irradiation (reflection mode, step size: 0.02°/step, 1 s/step). Scanning electron microscopy (SEM) was conducted using a ZEISS Merlin Compact Field Emission Scanning Electron Microscope with an acceleration voltage of 5 kV.

Transmission electron microscopy (TEM) was taken from a Tecani-G2 T20 and F20 operating at an acceleration voltage of 200 kV. The samples for TEM measurements were prepared by dropping ethanol dispersion of nanocrystals onto carbon-coated copper grids by using pipettes, and naturally dried under ambient condition.

The soft X-ray absorption spectroscopy data at the Ni and Fe L-edges as well as O and N K-edges were collected at room temperature at the BL11A beamline of National Synchrotron Radiation Research Center (NSRRC). All the spectra were recorded in the total electron yield (TEY) mode with the photon energy resolution of about 0.3 eV.

All the XAFS data were collected at room temperature at BL14W1 beamline of Shanghai Synchrotron Radiation Facility (SSRF). Monochromatized X-ray beam was provided by a double-crystal Si(111) monochromator, and then the rejection of higher harmonics was achieved by a pair of Rh-coated mirrors. The photon energies were calibrated to the first inflection point of the K edge from Ni foil at 8333 eV or Fe foil at 7112 eV, respectively. The reference spectra were recorded in transmission mode, while all the spectra of samples (including *ex situ* and *operando*) were measured in fluorescence mode with a Lytle detector filled with Argon gas. The sample for *operando* measurement was CP treated after 50h at a constant current density of 20 mA cm^{-2} , and several potentials were applied to the working electrode in 1 M KOH solution. The acquired EXAFS data were normalized using the ATHENA module implemented in the IFEFFIT software packages. The k^3 -weighted $\chi(k)$ data of Ni K-edge and Fe K-edge were Fourier transformed to real (R) space using a hanning windows ($dk=1.0 \text{ \AA}^{-1}$) to separate the EXAFS contributions from different coordination shells. The ARTEMIS code was used to obtain the quantitative structural parameters by least-squares fitting. The detailed information of *operando* XAS can be seen in Supporting Information (Figure S22 and S23).

Computational Methods. All spin-polarized density functional theory (DFT) calculations were carried out using the Vienna Ab-initio Simulation Package (VASP). The exchange-correlation potential was described by the generalized gradient approximation (GGA) in the form of the Perdew-Burke-Ernzerhof (PBE) exchange model. The Hubbard U ($U_{\text{eff}} = 4 \text{ eV}$) correction^{48,49} was used to treat 3d electrons of Ni. The electronic wave functions were expanded using a plane-wave basis set with an energy cutoff of 500 eV. The Brillouin zone was sampled in a Monkhorst-Pack $3 \times 3 \times 2$ k-points mesh for structural optimization and $6 \times 6 \times 4$ for static calculations. All the structures were fully relaxed until reaching a force threshold of 0.01 eV $\text{\AA}^{-1}$. For more details see Supporting Information (Figure S24).

ASSOCIATED CONTENT

Supporting Information.

The Supporting Information is available free of charge on the ACS Publications website at DOI:

Additional spectroscopic characterizations, data fitting results and computational methods.

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Notes

□ Xiaozhi Su, Yu Wang and Jing Zhou contributed equally to this work.
The authors declare no competing financial interest.

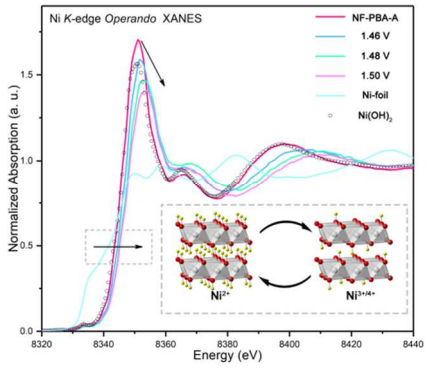
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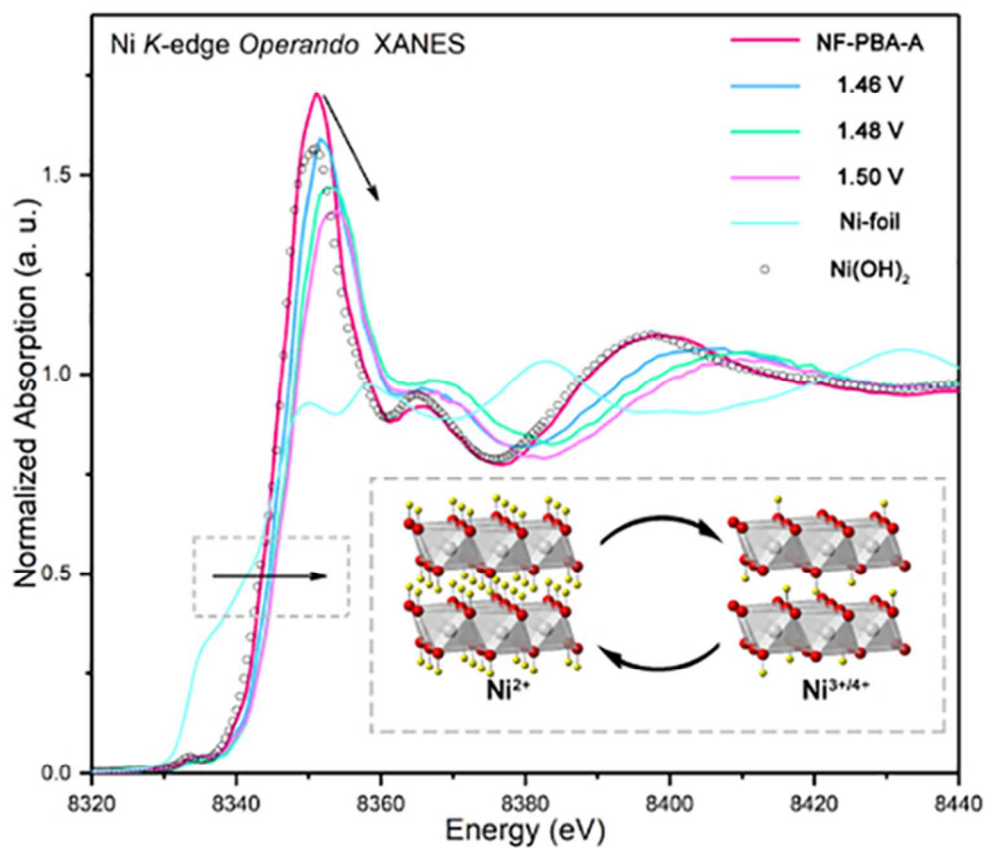
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